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Magnetostratigraphy across the end-Permian mass extinction event from the Meishan sections, southeastern China

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ABSTRACT

The end-Permian mass extinction (EPME) has been recorded as the most severe biodiversity crisis in Earth's history, although the timing of the marine and terrestrial extinctions remains debatable. We present a new high-resolution magnetostratigraphic succession across the EPME and the Permian-Triassic boundary (PTB) from the Meishan sections in southeastern China, which contain the global boundary stratotype section and point (GSSP) for the base of the Triassic (also the Induan Stage) and the base of the Changhsingian Stage. We identified five normal and five reverse magnetozones, including MS1n to MS5n and MS1r to MS5r, from oldest to youngest, in the Changhsingian and Induan Stages. The Induan Stage was determined to consist of two polarity intervals, where the upper one is reverse (MS5r), and the lower one is normal (MS5n). The Changhsingian Stage is dominated by normal polarity, intercalated with four short-term reverse magnetozones (MS1r to MS4r). Consequently, the PTB and the Wuchiapingian-Changhsingian boundary are clearly located in MS5n and MS1n, respectively. These new magnetostratigraphic results provide a potential reference geomagnetic polarity pattern with which to refine the geomagnetic polarity time scale for the EPME and the Permian-Triassic transition.

INTRODUCTION

The end-Permian mass extinction (EPME) across the Permian-Triassic boundary (PTB) has been universally documented as the most severe extinction event (Erwin, 1994) in the Phanerozoic, which eliminated over 81% of all marine species (Fan et al., 2020) and severely impacted the terrestrial ecosystem (Fielding et al., 2019; Gastaldo et al., 2020; Viglietti et al., 2021). High-precision chemical abrasion–isotope dilution–thermal ionization mass spectrometry (CA-ID-TIMS) U-Pb zircon geochronology (S.Z. Shen et al., 2011; Burgess et al., 2014), multiple geochemical profiles (Xie et al., 2007; Y. Shen et al., 2011; Shen et al., 2013; Liu et al., 2017), cyclostratigraphy (Wu et al., 2013; Li et al., 2016), and high-resolution biostratigraphy (Yuan et al., 2014; Brosse et al., 2016) have provided useful correlation tools for the EPME and PTB in marine sequences. Nonetheless, due to the different proxies and fossil markers for correlation, it remains debatable whether the timing of the EPME in marine and terrestrial sequences is consistent (Ward et al., 2005; Gastaldo et al., 2020; Viglietti et al., 2021). Magnetostratigraphy can provide a robust tool for correlating worldwide sedimentary strata in association with precise U-Pb ages and facilitate assessment of the timing and consistency of the EPME and PTB between marine and terrestrial ecosystems. However, high-resolution and reliable magnetostratigraphic records across the EPME and PTB are limited.

The renowned Meishan sections, located in southeast China (Figs. 1A and 1B), preserve continuous paleontologic and geologic records of the EPME and PTB in a marine setting. Consequently, the sections have been defined as the global stratotype section and point (GSSP) for the PTB (Yin et al., 2001) and the base of the Changhsingian Stage (Fig. 1C; Jin et al., 2006). The marine sequence is ascendingly composed of the Late Permian Lungtan Formation, the Changhsing Formation, and the Early Triassic Yinkeng Formation (Fig. 1C). The upper part of the Lungtan Formation is characterized by silty mudstone interbedded with thin-bedded calcareous sandstone. The Changhsing Formation (~45 m in thickness) mainly consists of thin- to medium-bedded limestones. The overlying Yinkeng Formation is composed of dark mudstone, interbedded with thin-bedded limestone and argillaceous

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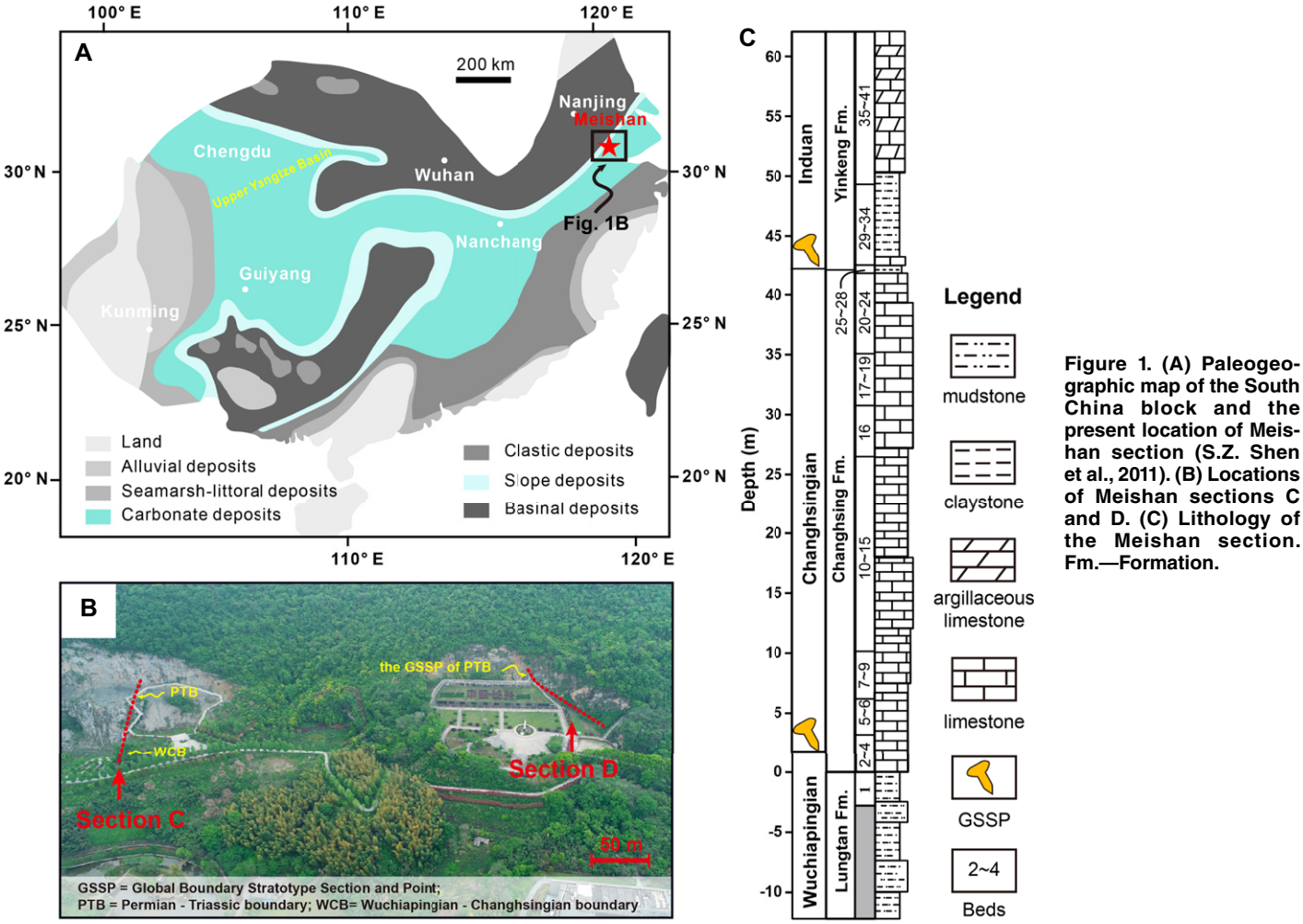


Figure 1. (A) Paleogeographic map of the South China block and the present location of Meishan section (S.Z. Shen et al., 2011). (B) Locations of Meishan sections C and D. (C) Lithology of the Meishan section. Fm.—Formation.

limestone (see the Supplemental Material[1] for detailed lithology).

Abundant fossils, especially conodonts and ammonoids, in these sections are well preserved and have been studied for biostratigraphic correlations (Yuan et al., 2014). In addition, large negative excursions of $\delta^{13}C_{carb}$, $\delta^{13}C_{org}$ (carb—carbonate; org—organic) (S.Z. Shen et al., 2011; Burgess et al., 2014), $\delta^{18}O_{apatite}$ (Joachimski et al., 2012; Chen et al., 2016), $\delta^{7}Li$ (Sun et al., 2018), $\delta^{66}Zn$ (Liu et al., 2017), and multiple sulfur isotopes (Y. Shen et al., 2011) have been established across the EPME interval. The EPME at the Meishan sections was precisely calibrated within an interval bracketed by two ash beds with a duration of 61 ± 48 k.y. (Burgess et al., 2014). However, its correlation with any form of interpreted terrestrial EPME has long been hampered; due in part because reliable magnetostratigraphy is lacking due to the extremely weak remanent magnetization of limestones (10^{-6}–10^{-5} A/m) at the Meishan sections.

In this study, we collected 649 paleomagnetic samples from the Changhsing and Yinkeng Formations from Meishan sections C and D (31°4.644′N, 119°42.510′E), ~0.4 km apart. All samples were thermally stepwise demagnetized by a custom-made PGL-100 thermal demagnetizer (constructed by Beijing Paleomagnetism and Geochronology Laboratory, PGL) with a low residual field of <5 nT (Qin et al., 2020). Remanent magnetizations were measured using the highly sensitive RAPID 2G Enterprises cryogenic superconducting magnetometer (Kirschvink et al., 2015), which is installed in a magnetically shielded room with a residual field of <300 nT. (For complete information on the sampling procedures, the paleomagnetic measurement methods, and data, see the Supplemental Material.) We then constructed a high-resolution magnetostratigraphic sequence from the Meishan sections. The new magnetostratigraphic data may enable us to correlate the marine-based EPME interval with marine and nonmarine (terrestrial) ecosystems.

RESULTS

Isothermal remanent magnetization (IRM) acquisition curves, back-field demagnetization curves, thermal demagnetization of three-component IRMs, and low-temperature measurements indicate that magnetite is the carrier of the remanent magnetization (Fig. S2; see the Supplemental Material for detailed descriptions of the rock magnetic procedures). Most specimens have two components of remanent magnetization (Fig. 2; Fig. S3). A low-temperature component (LTC) of remanent magnetization was unblocked below ~260 °C with a mean direction (declination [*D*] and inclination [*I*] in geographic coordinates [g]) of $D_g = 359.5°$, $I_g = +47.0°$, Fisher precision parameter $k = 25.4$, and confidence limit in geographic coordinates $\alpha_{95g} = 1.1°$ ($n = 643$ specimens), which is close to the present dipole field (PDF, $D = 354.2°$, $I = +47.2°$; Fig. S4A). A high-temperature component (HTC), calculated based on data obtained above ~260 °C, was successfully isolated and considered to be the characteristic remanent magnetization (ChRM; Fig. 2; Fig. S3). A Fisher averaged normal polarity direction in stratigraphic coordinates (s) of $D_s = 43.6°$, $I_s = +35.4°$, $k = 16.4$, and

[1]Supplemental Material. Data, details of experimental procedures, and supplemental figures and interpretations. Please visit https://doi.org/10.1130/GEOL.S.14810547 to access the supplemental material, and contact editing@geosociety.org with any questions.

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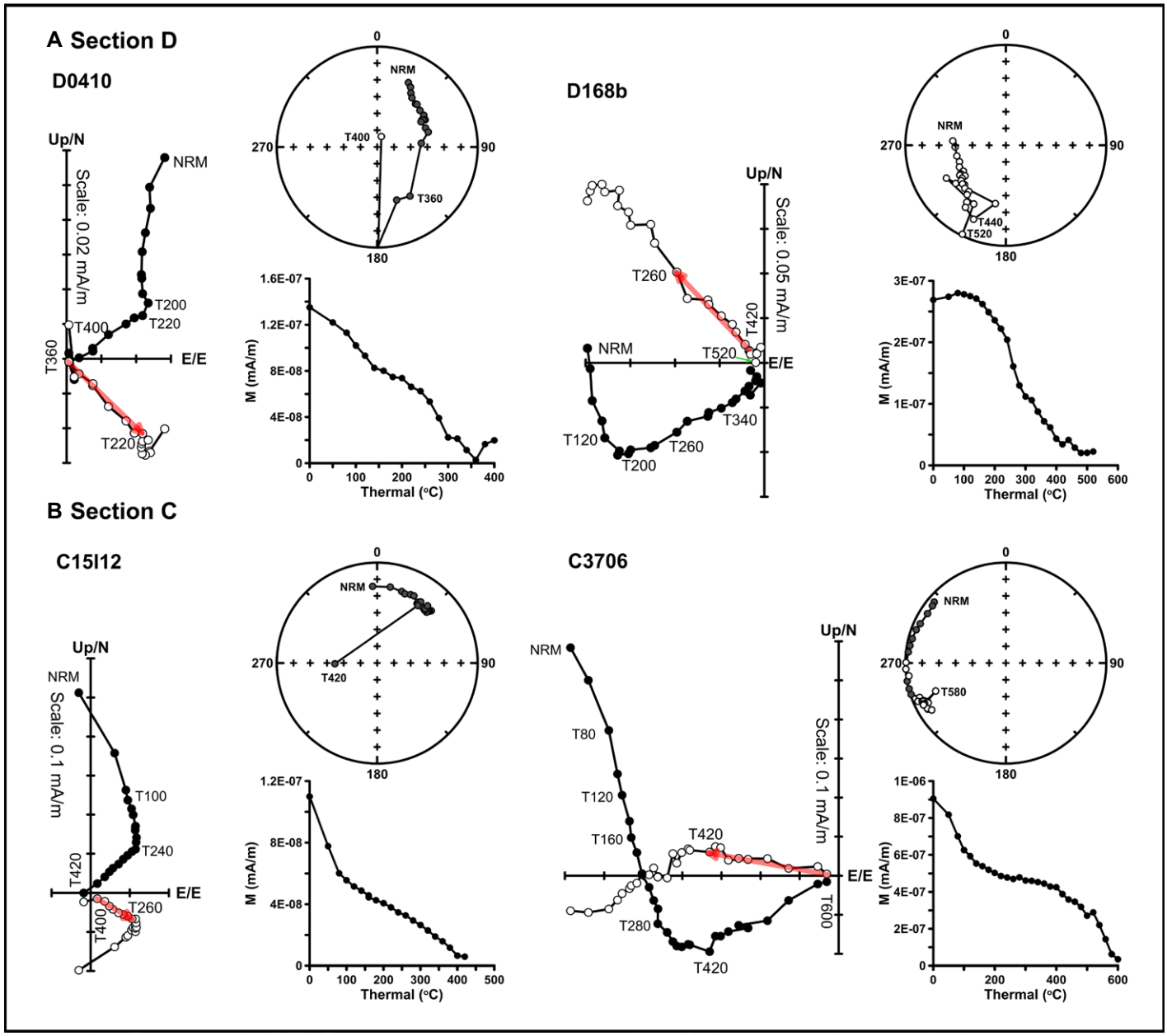


Figure 2. Thermal demagnetization results presented in stratigraphic coordinates. (A) Meishan (southeastern China) section D. (B) Meishan section C. Solid (open) circles denote horizontal (vertical) projection. Red arrows show characteristic remanent magnetization (ChRM) vector directions. NRM—natural remanent magnetization.

$\alpha_{95s} = 1.8°$ ($n = 381$ specimens) and an averaged reverse polarity direction of $D_s = 236.9°$, $I_s = -13.6°$, $k = 8.5$, and $\alpha_{95s} = 7.1°$ ($n = 54$ specimens) were calculated after tilting correction (bedding strike/dip = 248°/40° in section C and 253°/38° in section D, respectively; Fig. S4). These mean directions corresponded to a paleopole (long—longitude; lat—latitude) of Plong.$_s = 215.3°$, Plat.$_s = +47.4°$, A_{95s} (radius of the 95% confidence circle about the calculated mean pole) = 1.6°, and $k = 18.1$, which is consistent with other late Permian – Early Triassic poles for South China (Table S1). The fold test (Tauxe and Watson, 1994; see also Fig. S5) was clearly positive. These results indicate that the ChRM components were acquired during limestone deposition (see the Supplemental Material for detailed information on the fold test). These paleomagnetic results also show a *Q*-factor (quality index) of 7 according to the seven reliability criteria of Van der Voo (1990).

The ChRM direction for each specimen was calculated to a virtual geomagnetic pole (VGP) with latitudes based on the latest Permian to earliest Triassic paleomagnetic pole for the South China block (Table S1), referred to as rVGP latitude, which was used to establish the magnetostratigraphy. The results from three or more consecutive specimens of the same polarity were defined as a magnetozone (illustrated as full bars in Figs. 3 and 4), and those from only one or two consecutive specimens were defined as submagnetozones (illustrated as half bars in Figs. 3 and 4). The magnetozones were labeled MS for the Meishan sections. We delineated seven magnetozones and one sub-magnetozone at Meishan section C (Fig. 3A). Specifically, the lower part of Meishan section C was characterized by two thick normal magnetozones and two thin reverse magnetozones. The upper part was characterized by two thick normal magnetozones and one thick reverse magnetozone, with a reverse sub-magnetozone inserted in the lower part of the thick normal polarity interval. At Meishan section D, nine magnetozones were identified (Fig. 3B). There were two thick normal magnetozones and two thin reverse magnetozones in the lower part of the section. The upper part of

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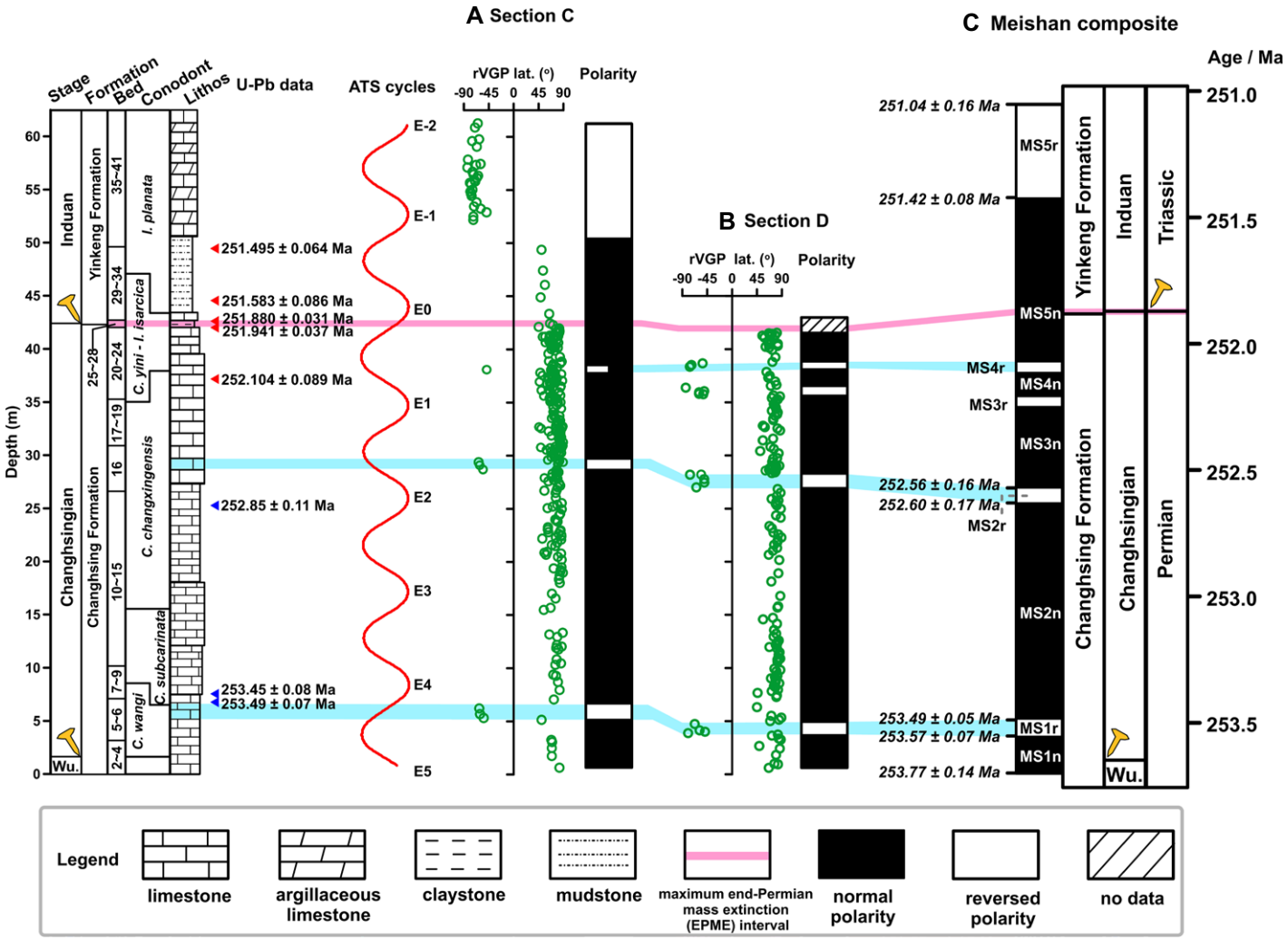


Figure 3. Magnetostratigraphic results of the Meishan (southeastern China) sections. (A) Meishan section C. (B) Meishan section D. (C) Composited magnetozones. Conodont zones are modified from Yuan et al. (2014). U-Pb data (blue and red triangles) are from S.Z. Shen et al. (2011) and Burgess et al. (2014), respectively. Astronomical time scale (ATS) is from Wu et al. (2013). NRM—natural remanent magnetization; rVGP—rotated virtual geomagnetic pole; Wu. —Wuchiapingian; *C.*—*Clarkina*; *I.*—*Isarcicella*.

this section was dominated by three relatively thick normal magnetozones intercalated with two thin reverse magnetozones. Combining Meishan sections C and D together, 10 magnetic polarity intervals (MS1n to MS5n, and MS1r to MS5r) were defined from the Changhsingian Stage to Induan Stage (Fig. 3C).

The lower Changhsingian Stage was characterized by two normal (MS1n and MS2n) magnetozones intercalated with two thin reverse (MS1r and MS2r) zones, with the distinctively thick MS2n spanning the *Clarkina subcarinata* conodont zone to the middle of the *C. changxingensis* zone. The upper Changhsingian Stage is characterized by three normal (MS3n − MS5n) magnetozones intercalated with two reverse (MS3r − MS4r) zones, while MS3r and MS4r are not well delineated in section C. The relatively thick MS5n begins somewhat above the top of the *C. changxingensis* zone, ranging from the upper part of the Changhsingian Stage to the lower part of the Induan Stage. Therefore, it is clear that the EPME and PTB occurred in the lower part of MS5n. At Meishan section C, the exposed interval of the Induan Stage is defined by a normal magnetozone (MS5n) and a younger reverse magnetozone (MS5r).

DISCUSSION

Over the past two decades, many magnetostratigraphic investigations have focused on the PTB interval (Hounslow and Balabanov, 2018). However, high-quality data are scarce, and the correlations between marine and terrestrial sequences are challenging (Table S5). At the Meishan sections, we established five thick normal magnetozones (MS1n − MS5n) and five reverse magnetozones (MS1r − MS5r) from the Changhsingian Stage to lower part of the Induan Stage (Fig. 3). To refine the age and duration of each magnetozone, we constructed a Bayesian age-depth model using the Bchron software package (Fig. S8; Parnell et al., 2008) based on the weighted mean radioisotope age estimates and their relative stratigraphic positions from the Meishan sections (Fig. 3C; S.Z. Shen et al., 2011; Burgess et al., 2014). The astronomical time scale (ATS) with a dominant 405 k.y. cycle was also considered (Wu et al., 2013). The lower part of Changhsingian Stage is characterized by a thick MS2n magnetozone, which is estimated to be between 253.49 ± 0.05 Ma and 252.60 ± 0.17 Ma in age. The MS5n magnetozone spans the upper Changhsingian Stage to the lower Induan Stage, beginning at 252.09 ± 0.06 Ma and terminating at 251.42 ± 0.08 Ma. The PTB, defined by the first occurrence of the conodont *Hindeodus parvus* (at the middle of Bed 27; Yin et al., 2001), lies in the lower part of MS5n. The EPME also falls within MS5n, which has an age duration between 251.941 ± 0.037 Ma (=Bed 25) and 251.880 ± 0.031 Ma (=Bed 28), starting with a sharp excursion of carbon isotopes (Burgess et al., 2014; S.Z. Shen et al., 2011). Thus, this robust magnetostratigraphic framework for the PTB and the EPME at the Meishan sections,

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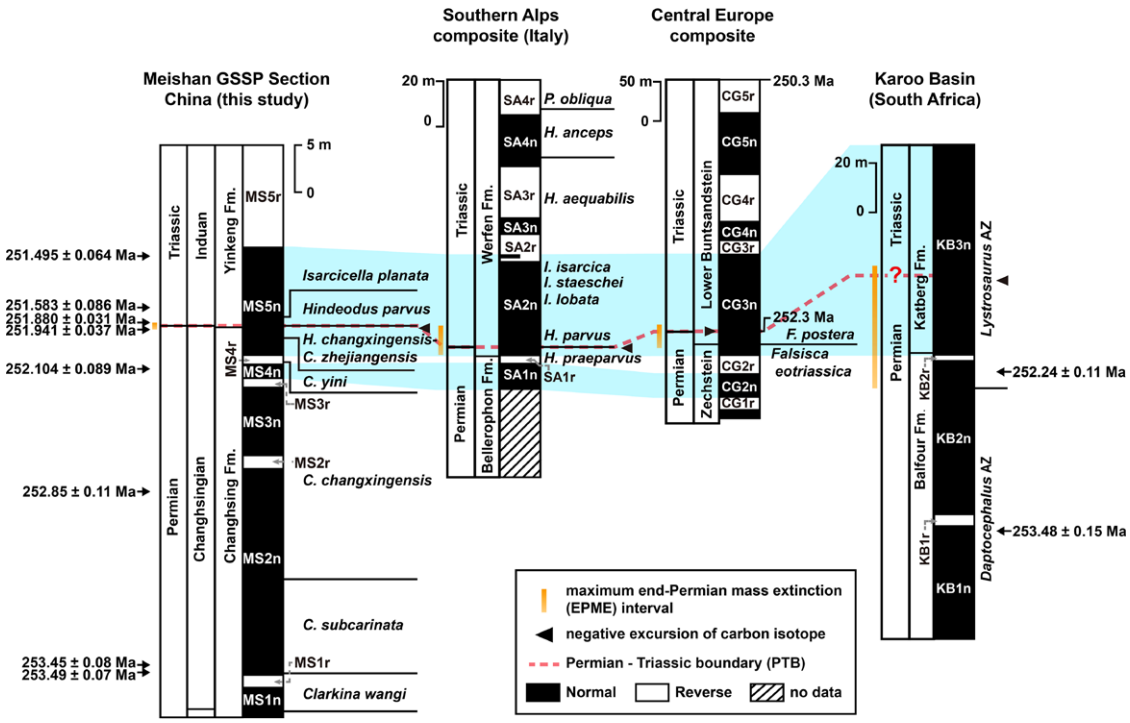


Figure 4. Biomagnetostratigraphic comparison of Meishan (southeastern China) sections with the southern Alps (Italy) (Scholger et al., 2000; Brandner et al., 2009), the Central European Basin (Szurlies et al., 2012), and the Karoo Basin (South Africa) (Gastaldo et al., 2018, 2020). U-Pb data in Meishan sections are from S.Z. Shen et al. (2011) and Burgess et al. (2014). Tops of MS4r, SA1r, CG2r, and KB2r magnetozones are set as horizontal isochrons. Prefixes SA and KB are used to label magnetozones in the southern Alps and the Karoo Basin, respectively. GSSP—global stratotype section and point; AZ—assemblage zone; *C.*—*Clarkina*; *F.*—*Falsisca*; *H.*—*Hindeodus*; *I.*—*Isarcicella*; *P.*—*Pachycladina*.

along with geochronologic, biostratigraphic, and chemostratigraphic data, forms a useful reference, allowing correlation with global data.

For the southern Alps (Italy) composite section, the first occurrence of *H. parvus* and the globally recognized negative carbon isotope excursion are both within a thick normal polarity magnetozone (SA2n) from the Tesero to Mazzin Members of the Werfen Formation, which overlies a thin reverse magnetozone (SA1r) in the uppermost part of the Bellerophon Formation (Fig. 4; Brandner et al., 2009; Scholger et al., 2000). Thus, SA2n and SA1r are generally correlated with MS5n and MS4r, respectively. In the Central European Basin, the PTB is expressed by the last occurrence of conchostracan *Falsisca postera* associated with a negative carbon isotope excursion that is slightly above the Zechstein-Buntsandstein boundary (Bachmann and Kozur, 2004). The boundary is related to a remarkable palynofloral turnover with a strong increase in humidity and freshwater discharge, which constitute proxies for the EPME in the Central European Basin (Kürschner and Herngreen, 2010), falling within the lower part of the thick normal magnetozone (CG3n) (Szurlies et al., 2012; Szurlies, 2013). Thus, we hypothesize that the CG3n and CG2n magnetozones are equivalent to MS5n and MS4n in the Meishan sections, respectively (Fig. 4).

In the Karoo Basin (South Africa), the turnover from the vertebrate *Daptocephalus* assemblage zone to the *Lystrosaurus* assemblage zone has been recently recognized as the onset of the EPME, followed by rapid faunal turnover in the uppermost part of the Balfour Formation (Palingkloof Member; Gastaldo et al., 2020; Viglietti et al., 2021). Based on magnetic polarity stratigraphy, a thick normal magnetozone is preceded by a narrow reverse polarity interval (KB2r) in the topmost part of the Balfour Formation (Gastaldo et al., 2018). The magnetozone KB2r could be correlated to MS4r in the Meishan sections (Fig. 4) based on a high-precision U-Pb age of 252.24 ± 0.11 Ma (Gastaldo et al., 2020). Thus, the marine EPME in magnetozone MS5n and the PTB at the Meishan sections lie slightly above the lithological boundary between the Balfour Formation and the Katberg Formation and fall into the thick normal magnetozone (KB3n). This inconsistency in the EPME between the Karoo Basin and the Meishan sections could be due to the Signor-Lipps effect. The vertebrate fossil record of the Karoo Basin is much coarser than the marine record at the Meishan sections (Gastaldo et al., 2018; Viglietti et al., 2021). Nonetheless, the higher last occurrences of a dicynodontoid skull and the wood fossil *Agathoxylon africanum* and the persistence of the *Glossopteris* (seed ferns) flora suggest that the terrestrial EPME extends into the lower part of the Katberg Formation (Gastaldo et al., 2018, 2020; Viglietti et al., 2021). Although our new magnetostratigraphic data from the Meishan sections demonstrate that the EPME in the marine and terrestrial ecosystems was essentially isochronous, more precise correlation of the EPME and PTB between the terrestrial Karoo Basin and the marine Meishan sections needs further investigations.

CONCLUSIONS

Our magnetostratigraphic study provides a high-resolution geomagnetic polarity sequence, consisting of five normal magnetozones (MS1n – MS5n) and five reverse magnetozones (MS1r – MS5r), in the renowned Meishan sections that contain the entire Changhsingian Stage, two GSSPs, and a record of the EPME. The Changhsingian Stage is dominated by normal polarity intercalated with four short-term reverse polarity intervals (MS1r to MS4r). The investigated lower part of the Induan Stage was determined to consist of two polarity intervals: an upper reverse magnetozone (MS5r) and a lower normal (MS5n) magnetozone. A thin reverse magnetozone (MS4r) and a thick normal magnetozone (MS5n) spanning the uppermost Changhsingian Stage to the lower Induan Stage, with a duration of ~0.7 m.y., can be regarded as a distinct marker for terrestrial-marine correlation. Consequently, we infer that both the PTB and EPME intervals are in MS5n.

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